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Digital marketing adoption and success for small businesses

The application of the do-it-yourself and technology acceptance models

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Abstract

Purpose – This paper aims to examine small business' participation in digital marketing and to integrate the do-it-yourself (DIY) behavior model and technology acceptance model (TAM) so as to explore the motivations and expected outcomes of such participation.

Design/methodology/approach – Data from 250 small business owners/managers who do their own digital promotion are collected through an online survey. Structural equation modeling is used to analyze the relationships between the models.

Findings – The results contribute to the understanding of small business' digital marketing behavior by finding support for the idea that the technological benefits may not be the only motivators for small business owner/managers who undertake digital marketing. Moreover, and perhaps more importantly, the authors find that the DIY behavior model applies to small business owner/managers who must perform tasks that require specialized knowledge.

Research limitations/implications – The limitations of this research are that the motivations to undertake digital marketing are limited to those contained in the DIY and TAM models, and the sample may not be representative of all owners and managers who perform digital marketing for their small businesses. Therefore, future research is needed to determine if further motivations to conduct digital marketing exist and whether other samples produce the same interpretations.

Originality/value – This study presents empirical evidence supporting the application of the DIY model to a context outside of home-repair and extends the understanding of digital footprint differences between large and small businesses.

Keywords Small business, Digital marketing, DIY, TAM, Motivation

Paper type Research paper

Introduction

Digital marketing can be defined as the promotion of goods and services "using digital technologies, mainly on the Internet, but also including mobile phones, display advertising, and any other digital medium" (https://en.wikipedia.org/wiki/Digital_marketing) or, similarly, "the practice of promoting products and services using digital distribution channels via computer,

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mobile phones, smart phones, or other digital devices" (Smith, 2012, p. 86). The use of digital channels has transformed the way marketers communicate with today's consumers. A considerable portion of the world's consumers own and use computers and/or mobile devices, which contributes to the tremendous growth of digital ad spending. Marketers quickly recognized the benefits of social networks such as Facebook, YouTube, Twitter, Instagram, Snapchat, Pinterest and LinkedIn for communications and spent $51.3bn on global social network advertising in 2017, a 55.4 per cent increase from 2016 (Cooper, 2018). The amount spent on digital ads is expected to increase by 17.7 per cent in 2018 and comprise $273bn (44 per cent) of the $629bn spent on advertising globally (McNair, 2018). Mobile ad spending grew 39 per cent in 2017 and is forecast to grow another 27 per cent and constitute 55 per cent of all digital ad spending in 2018 (MAGNA Global). The increasing concentration of advertising dollars is compelling evidence of digital marketing's effectiveness for reaching target markets and achieving growth objectives that include increased sales, brand awareness, customer engagement, lead generation and reduced customer acquisition and support costs (Labrecque *et al.*, 2013; Lamberton and Stephen, 2016; Tuten and Solomon, 2015).

Despite the known benefits of digital promotions, little is known about digital marketing by small businesses because the majority of the digital marketing literature focuses on large businesses and organizations (Celuch and Murphy, 2010; Järvinen *et al.*, 2012; Michaelidou *et al.*, 2011). Large businesses are expected to have websites that also are mobile enabled, and they can hire outside experts to manage search engine optimization projects and social media marketing firms to implement and run social media campaigns, whereas small business owners "develop, change, and evolve their marketing activity intelligence through social media use" (Atanassova and Clark, 2015, p. 163). Both the practitioner and academic literature assume that businesses outsource some or all of the digital marketing functions (Edelman, 2010; Leeflang *et al.*, 2014; Montalvo, 2011), yet 55 per cent of small businesses in the USA do not have a webpage (Pisani, 2014), largely due to financial constraints (Chaffey, 2010). The amount of investment for digital marketing is dependent on the firm's existing marketing strategies and expectations for success (Reichheld and Schefter, 2000). Small businesses likely would benefit from participating in and developing a digital marketing strategy, and the lack of such a strategy broadens the performance gap between large and small businesses due to reduced opportunities to reach target markets and stimulate sales growth. Thus, compared to large businesses, small businesses have different digital footprints and technology adoption speeds (Harrigan *et al.*, 2011; Nguyen *et al.*, 2015), which calls for specific research of their digital marketing use.

The purpose of this study is to explore small businesses' use of digital marketing by investigating the motivations to participate in the activity. Previous research examines the motivations for technology adoption at large firms, but there are alternative factors that could explain whether a technology is adopted by small business owners and managers. The willingness to adopt technology traditionally is explained by the technology acceptance model (TAM) (Davis, 1989; Venkatesh *et al.*, 2003), which typically is applied to consumers. However, the TAM has been applied to businesses through studies on, for example, the adoption of social media sites for marketing (Lacka and Chong, 2016; Michaelidou *et al.*, 2011; Siamagka *et al.*, 2015), the firm's ability to sense and respond to 'technological opportunism' (Srinivasan *et al.*, 2002), and the proactive adoption of functional, inter-firm technologies such as radio frequency identification (RFID), Global Positioning Systems (GPS) and other supply chain technologies (Asare *et al.*, 2016). Studies of the motivations to adopt technology at the firm level include examples such as IT readiness (Qu and Wang, 2011) and the coercive power one firm has over another (Zhang and Dhaliwal, 2009). The influences of technology adoption in the small business environment are less complex, and

include factors such as resource limitations, risk, procedural complexity, and technical challenges (Alam, 2009; Dahnil *et al.*, 2014; Gilmore *et al.*, 2007; Yeung *et al.*, 2003).

Because the decision-making processes of small business owners and managers reflect those of individual decision-making behaviors (Dahnil *et al.*, 2014), we combine the TAM with a second model to explain the adoption of digital marketing by small businesses. Specifically, we simultaneously consider the TAM (Davis, 1989) and the do-it-yourself (DIY) behavior model (Wolf and McQuitty, 2013), which was developed in the context of consumers' motivations to DIY (create products themselves) and the associated outcomes of such behaviors. The rationale for combining these models stems from similarities between individuals who undertake DIY activities (or DIYers) and small business owners and managers. As with DIYers, small business owner/managers typically are constrained by their financial resources, and may perceive market solutions as either unavailable or lacking quality.

Extending the DIY behavior model to small business owners' and managers' use of digital marketing activities is relevant because owners/managers often must take on a variety of business activities with little or no training. The performance of a small business is highly dependent on the abilities of owners and managers to carry out such tasks successfully (McGowan and Durkin, 2002). Keeping business activities in-house is preferable because outsourcing can be costly and may not provide the service needed or with the desired quality. These are the same motivations for DIY behaviors that Wolf and McQuitty (2013) study, which suggests that evaluating their model in the small business context is appropriate.

The small business context

There are a variety of definitions of what constitutes a small business. The World Bank categorizes the size of firms by number of employees, and describes firms with 1-9 employees as *micro*, and firms with 10-49 (or 10-99, depending on the country) as *small* (Kushnir *et al.*, 2010). According to the US Small Business Administration size standards, small firms are defined as having fewer than 500 employees (USA Small Business Administration, 2016); however, small business administration (SBA) classifications can vary across loan programs, industry, and annual revenue. For example, a small business classification can be assigned to firms with fewer than 100 employees in the retail sector and as many as 1,500 employees in the information, publishing, and manufacturing sector (USA Small Business Administration, 2016). Because of the variation in employee number definitions across industries and to avoid classification overlaps, for the purposes of our research we define small businesses as having fewer than 50 employees. Such businesses account for nearly half (48 per cent) of US GDP and employ 27.8 per cent of all workers (USA Small Business Administration, 2014). Businesses with fewer than 50 employees account for nearly 60 per cent of global GDP, with total employee count equal to the world's larger corporations (Kushnir *et al.*, 2010).

Small businesses are likely to have an owner or manager whose responsibilities could include, among others, the undertaking or overseeing of electronic marketing activities (Nguyen *et al.*, 2015; Rogers, 2004). Small businesses tend to struggle with limited resources (temporal, financial, technical, and managerial), which plays a role in the uncertainty regarding the use of technology (Bhagwat and Sharma, 2007). Previous research on information technology (IT) acceptance using similar small business contexts (i.e. <50 employees) finds that the low IT adoption rate and the high rate of business failure for small businesses can be attributed to weaknesses in the "organization, internal IT resources, external IT consultants, supplier relations, and customer relations" (Nguyen *et al.*, 2015,

p. 208). Although access to digital marketing media is free and open, an entire service industry has emerged where professionals specialize in search engine optimization (SEO), ecommerce and social media management systems, and can perform these activities for other businesses. However, the implementation of digital marketing by small businesses typically is done in an experiential or "learn by doing it yourself" method (Cope, 2005). Through the process of self-learning, the small business marketer develops a sense of control and ensures that marketing efforts enhance relationships with customers (Jones *et al.*, 2013; Malthouse *et al.*, 2013).

Theory

There are advantages and disadvantages to using the Internet for marketing purposes. Web 2.0 and interactive technologies facilitate the two-way marketing communications that build brands, increase customer loyalty and improve business performance (Bacile *et al.*, 2014; Chawanuan *et al.*, 2015; Prahalad and Ramaswamy, 2004). However, with online interactions comes an obligation for transparency from marketers (Bacile *et al.*, 2014). Some businesses perceive an element of risk associated with transparency and therefore view online interactions negatively. Interconnectivity requires that organizations ensure security and privacy, while preventing negative outcomes for both the customer and the organization (Limbu *et al.*, 2011, 2012).

On the other hand, when marketers are open to "consumer involvement in co-producing the communication process" (Bacile *et al.*, 2014, p. 28), the communication becomes more effective and valuable to both the consumer and the business. Co-production of marketing communications requires some transfer of control that allows consumers to have input on the frequency, time of day, and relevance of the marketing communications viewed on their personal mobile devices. Furthermore, although successful small businesses can develop a competitive advantage over large companies through the personal, face-to-face relationships with customers (Harrigan *et al.*, 2011), transparency obligations, security risks related to integrative online marketing, and the creation and implementation of digital marketing campaign elements present challenges for small business owners and managers.

Acquiring the expertise necessary to engage in digital marketing is viewed as a hurdle for small businesses (Järvinen *et al.*, 2012). There are a variety of skills needed to implement digital marketing, which can be categorized as external (technology) or internal (objectives and campaign outcomes). The technical tools used for digital marketing facilitate the creation and maintenance of websites, social media sites, writing and posting content (blogs, photos, videos, customer responses), managing third party application systems such as Wordpress, search engine optimization (SEO), and tracking performance indicators through analytics (Google, Facebook, etc.). Examples of digital marketing goals include increasing customer engagement (comments, reviews, recommendations), awareness (shares, clicks, likes, views), increasing sales, adding value (as a subject matter expert), loyalty and providing opportunities for customer co-creation (new product development) (Smith, 2012; Tuten and Solomon, 2015; Truong and Simmons, 2010).

To explore small business owners' and managers' motivations to undertake digital marketing, we apply two existing models to study the antecedents and outcomes of participating in digital marketing. Both of these models, the TAM (Davis, 1989; Venkatesh *et al.*, 2003) and the DIY behavior model (Wolf and McQuitty, 2013), describe possible motivations to undertake digital marketing and the outcomes from such activity. We describe these models and their relationships for small business owners and managers implementing digital marketing.

The technology acceptance model

The first model that we associate with the implementation of digital marketing is the TAM (Itani *et al.*, 2017; Jelinek *et al.*, 2006; Lacka and Chong, 2016), which proposes that an individual's perceptions of a technology's *ease of use* and its *usefulness* are the determinants of intentions to adopt the technology and actual adoption behavior (Davis, 1989; Venkatesh *et al.*, 2003). A principle underlying the TAM is that the easier a technology is to use, then the more beneficial it is to the user (Venkatesh and Davis, 2000). Examples of marketing related topics that apply the TAM are self-serve technology (Chowdhury *et al.*, 2014; Dabholkar and Bagozzi, 2002), social media adoption (Veldman *et al.*, 2015); mobile CRM technology (Rodriguez and Trainor, 2016), sales force automation tools (Homburg *et al.*, 2010), and ecommerce (Ashraf *et al.*, 2014). Technology ease of use and usefulness also are associated with, for example, post-use evaluations (Kim and Forsythe, 2008), revisit intentions (Reynolds and Ruiz de Maya, 2013), and attitudes (Klein, 2003; Kulviwat *et al.*, 2014). The TAM does not appear to have been studied previously in relation to the motivations and expected outcomes of digital marketing by small business owners and managers.

The do-it-yourself behavior model

Whereas the TAM evaluates the intentions to use a technology based on the *perceived ease of use* and the *usefulness* of the technology, the DIY behavior model (Wolf and McQuitty, 2013) evaluates motivations (the *economic benefits* and the *lack of quality and availability* in existing products) to undertake a DIY project. The DIY behavior model reflects the notion of *prosumption* (Toffler, 1980), which suggests that people increasingly will become engaged in producing the products they later consume (Kotler, 1986; Xie *et al.*, 2008). DIY activities are excellent examples of prosumption behaviors. Wolf and McQuitty (2013) finds that people evaluate marketplace factors such as the economic benefits, quality sought and product availability when considering the make-or-buy decisions to produce their own goods and services.

Wolf and McQuitty (2011) define DIY as behaviors "where individuals engage raw and semi-raw materials and component parts to produce, transform, or reconstruct material possessions, including those drawn from the natural environment (e.g. landscaping)" (p. 154). We acknowledge that this definition applies to a context in which materials are used, but believe that the term DIY has broader applications. For example, an online search reveals that the phrase "DIY" has extended from being associated exclusively with home improvement projects to nearly every realm of consumer culture, including such areas as music, arts and crafts, fashion, software engineering, and movie production.

A broader definition of DIY also suggests that the motivations and outcomes considered by Wolf and McQuitty (2013) may hold in other contexts, such as performing the activities necessary for running a small business. Wolf and McQuitty (2013, p. 198) state that:

> The physical and cognitive skills required by typical DIY activities extend the notion of value creation from how to use, maintain and repair [. . .] to the consumer's direct participation in the process of planning, designing, and constructing a product through self-effort.

Thus, the person considering the DIY activity searches for and obtains the physical and cognitive skills required to complete a DIY task.

It is important to recognize that the responsibilities associated with owning and running a small business effectively are DIY activities. Small business owners and managers typically are involved in planning, organizing and creating value, either on their own or by leading others. They take on financial risk and persist through uncertainty, often without

the aid of experts. Like many DIY activities, participating in digital marketing can be a laborious, time consuming, and expensive undertaking. It is this involvement and the requirement for the owner or manager's self-effort that makes the application of the DIY behavioral model valid in the small business context. Consequently, we apply the Wolf and McQuitty (2013) DIY behavior model to the participation in digital marketing by small business owners and managers.

Hypotheses development

We propose a model that combines the TAM and the DIY behavior models that, in this context, are used to capture small business owners' and managers' perceptions of using the technology associated with digital marketing. We use both models in an effort to broaden the range of motivations and expected outcomes of small businesses' use of digital marketing, and because conclusions drawn only from the TAM may produce an incomplete picture (Richard *et al.*, 2007). Moreover, small businesses face greater challenges (such as limited temporal, financial, technical, and managerial resources) than larger businesses with the creation and implementation of digital marketing (Harrigan *et al.*, 2011). Small businesses also have challenges for creating customer value with limited resources and inspiring managers to achieve intrinsically motivated outcomes, such as fulfillment and a sense of accomplishment through learning (Bontis *et al.*, 2002; Real *et al.*, 2014; Sinkula *et al.*, 1997), so investigating the perceptions of participating in digital marketing using both the TAM and the DIY behavior model has value.

Technology acceptance model hypotheses – antecedent factors

TAM is rooted in acceptance or behavioral theory. Fishbein and Ajzen's (1975) theory of reasoned action (TRA) offers a fundamental model for predicting human behavior given an individual's *attitude* and a *subjective norm*. The theory of planned behavior (TPB) (Ajzen, 1988), an extension of the TRA, introduced a *control beliefs* factor that reflects perceptions about factors that could affect a planned behavior. The TRA and TPB models have been applied in business areas such as marketing, accounting, information systems, and management.

The TAM is distinct in that it applies the TRA and TPB models with a focus on predicting information technology acceptance and usage. The four TAM constructs that are applied in this study are *perceived usefulness, perceived ease of use, intentions*, and *actual technology use* (behavior). The definition of perceived usefulness is "[. . .] the extent to which a person believes that using the system will enhance his/her job performance" (Venkatesh and Davis, 2000, p. 187). The definition of perceived ease of use is "[. . .] the degree to which a person believes that using a particular system would be free of effort" (Davis, 1989, p. 320). Derived from the TRA (Fishbein and Ajzen, 1975), the behavioral intentions to use digital marketing reflect "a person's subjective probability that he/she will perform some behavior" (Fishbein and Ajzen, 1975, p. 288). The dependent variable, actual technology use, measures the frequency, duration, and intensity of interactions with a technology (Brown *et al.*, 2010; Venkatesh *et al.*, 2003).

Traditional methods of building relationships are face-to-face, and some managers are skeptical about the effectiveness of social media for connecting with customers (Cespedes, 2015; Lacka and Chong, 2016; Swani and Brown, 2011). Yet, other managers see the benefits of reviews and recommendations from actual customers, because "foot traffic to retail businesses is down 57 per cent in the past five years, but the value of each visit has tripled", (Capoccia and Forbes Technology

Council, 2018). The use of technology is not as much a deterrent to adopting digital marketing as knowing which sites to use and how to best use them (Lacka and Chong, 2016; Michaelidou *et al.*, 2011). Small business owners and managers can gain initial experiences with digital marketing through participation in personal social media accounts, and subsequently with accounts specifically for the small business. In our context, actual technology use means that the small business owner/manager is undertaking digital marketing as a DIY behavior. Therefore, and consistent with the TAM, it is hypothesized that:

H1a. The perceived ease of using digital marketing has a positive effect on small business owners/managers' intentions to use digital marketing.

H1b. The perceived usefulness of digital marketing has a positive effect on small business owners/managers' intentions to use digital marketing.

H1c. Intentions to use digital marketing are positively related to the use of digital marketing.

Technology acceptance model hypotheses – post-adoption factors

As the relative and absolute amount spent on digital marketing increases and soon is expected to surpass that of traditional communication mediums (eMarketer, 2016), it becomes increasingly important to explore post-adoption attitudes toward digital technologies. Technology adoption and innovation diffusion research examines post-adoption variables such as satisfaction, disenchantment and intentions to continue use (Bianchi and Andrews, 2012; Son and Han, 2011; Sun, 2013). The intentions to use technology influence actual behavior, and the process of implementing digital technology produces expectations, additional information and personal experiences with marketing tactics and technology use (Venkatesh and Morris, 2000). We predict that the convergence of perceived expectations, new information, and actual experience influences the small business owner/manager's decisions to continue use, modify use, or discontinue use of digital marketing.

Intentions to discontinue using technology refers to what Bhattacherjee (2001) calls the "acceptance-discontinuance anomaly" (p. 352), which involves the initial acceptance and trial stage of technology, moving through the expectation-vs-reality stage, and ending with a decision to discontinue or switch to another form of technology (e.g. from a GoDaddy website platform to a WordPress website platform). Intentions to discontinue are distinct from dissatisfaction because these intentions recognize that digital marketing media are dynamic with new products and service substitutions frequently available (Venkatesh and Davis, 2000), which can change perceptions of technology and technology use (Bacile and Goldsmith, 2011; Sun, 2013).

Satisfaction with technology use is derived from a perception that there is an acceptable gap between one's expectations and actual experience (Son and Han, 2011). Marketers who use multiple digital media are considered more satisfied than those who do not, and "satisfaction with technology may spur more usage" (Chuan-Fong and Venkatesh, 2004, p. 63). However, small businesses' use of technology for digital promotions is low (Pisani, 2014). Switching from one product to another can be attributed to negative experiences with the former product and positive features of an alternative product (Jones *et al.*, 2000). We therefore hypothesize that:

H2a The quantity of digital marketing activities is negatively related to intentions to discontinue use of digital marketing.

H2b. The quantity of digital marketing activities is positively related to satisfaction with the use of digital marketing.

Do-it-yourself behavior model hypotheses – antecedent factors

The idea of doing-it-yourself has been around for many years, but only relatively recently have marketers begun to explore the motivations and outcomes of DIY behaviors in a consumer context (Wolf and McQuitty, 2011). DIY gives consumers the ability to circumvent traditional markets with what effectively are make-or-buy decisions (Wolf and McQuitty, 2013), and we propose that small business owners and managers may follow the DIY behavior model by choosing to perform their own business activities, such as digital marketing. The motivations (antecedents) for DIY behaviors found in the marketplace include the *perceived economic benefit, lack of product quality*, and *product availability* (Wolf and McQuitty, 2013). In a small business context, owners and managers can evaluate the marketplace to determine whether the available offerings provide a desired solution, or do their own digital marketing when it is a more efficient use of resources than external providers.

The most obvious motivation for DIY behaviors is the perceived economic benefit, which relates to the need for economic gain or a utility for saving (Wolf and McQuitty, 2011, 2013). Seeking economic benefit is not purely derived from low income, because the decision to perform DIY activities for economic reasons also can stem from frugal behavior (Lastovicka and Joachimsthaler, 1988; Carson, 1985), a desire to simplify (Huneke, 2005), or to redistribute financial resources (Craig-Lee and Hill, 2002). Similarly, small businesses typically have budget constraints and must distribute financial resources to maximize their effect. Performing the necessary activities themselves gives small business owners and managers the ability to closely interact with their environments, which is linked to better performance in operative, strategic and financial flexibility, and allows for quick adjustments and efficient resource distribution (Verdu-Jover *et al.*, 2006).

A *lack of product quality* relates to poor goods and services from professionals (digital marketing services, in our small business context). Prior to viewing DIY as a possible response to a perceived lack of product quality, it was assumed that consumers dealt with these inadequacies through complaining behaviors (Tronvoll, 2012) such as negative word of mouth and boycotts (Brown and Beltramini, 1989). However, the academic literature on service quality failure and strategies to bypass such failure is limited (Flores and Primo, 2008), and research on the make-or-buy decision centers on businesses and not consumers, with topics such as vertical integration in manufacturing and the costs associated with service provider failure (Jayawardhena *et al.*, 2007; Zimmermann *et al.*, 2016). Because strategies for addressing a lack of quality from a small business owner/manager's perspective have not received much attention, we use the Wolf and McQuitty (2013) notion of DIY behavior as a possible response to a perceived lack of quality.

A *lack of product availability* is the third factor Wolf and McQuitty (2013) describe as a motivation to perform DIY behaviors. If an owner/manager has the perception that specific services for digital marketing are difficult to obtain in the market place, then small businesses may be more likely to perform such services themselves. For example, digital service providers may decline a task or contract if the job is perceived as too small or unprofitable, as could be the case for typical small businesses.

Thus, the Wolf and McQuitty (2013) DIY behavior model suggests three motivations for small business owner/managers to perform digital marketing activities themselves (the perceived economic benefit, lack of product quality, and lack of product availability), which gives them the opportunity to better respond to customers. Having control over the timing and content of posts, pictures, and videos, for examples, allows the small business owner/manager to design and control the tone of all digital marketing communications (Holliman and Rowley, 2014). Given the limited resources associated with small businesses, we expect that the following factors are of importance and hypothesize that:

H3a. The perceived economic benefits are positively related to a small business using digital marketing.

H3b. The perceived lack of product quality is positively related to a small business using digital marketing.

H3c. The perceived lack of customized product availability is positively related to a small business using digital marketing.

Do-it-yourself behavior model hypotheses – outcomes

In addition to studying motivations for DIY behaviors, Wolf and McQuitty (2013) considers several outcomes deriving from such behaviors. We follow this framework and use the higher-order outcomes developed by Kahle (1983), Rokeach (1973), Herche (1994), and Xie *et al.* (2008) to "focus on people's values towards life and the self" (Wolf and McQuitty, 2013, p. 198). The outcome variables relevant for DIY behaviors in the small business context are perceptions of *control*, *fun and excitement*, and *self-improvement*.

Outcomes such as control, fun and excitement and self-improvement are important for the internal function of the business (Speier and Venkatesh, 2002). A sense of control is important when one uses prior knowledge and contextual information to manage the environment. For small businesses using digital marketing, the sense of control construct suggests that owner/managers can use technology to achieve specific tasks that produce subjective feelings of being in command (Wen *et al.*, 2015). The sense of fun and excitement construct suggests that people who engage in DIY behaviors can obtain feelings of pleasure and entertainment, because the activity provides an enjoyable experience and therefore is actively sought. A sense of self-improvement occurs when people engaging in DIY behaviors test their knowledge and skills, which provides a platform for creative involvement when solving problems. Consequently, we hypothesize that small business owner/managers who participate in digital marketing can experience senses of control, fun and excitement, and self-improvement:

H4a. Participation in digital marketing by small business owners and managers is positively related to a sense of control.

H4b. Participation in digital marketing by small business owners and managers is positively related to a sense of fun and excitement.

H4c. Participation in digital marketing by small business owners and managers is positively related to a sense of self-improvement.

Figure 1 depicts the conceptual model that integrates the TAM and DIY behavior models. The motivations and outcomes from developing and implementing a digital marketing strategy are reflected by the hypotheses and are illustrated in Figure 1. The following sections describe a study capable of evaluating the hypotheses concerning the motivations for digital marketing and its outcomes.

Methodology

To evaluate the conceptual model in Figure 1 and test the hypotheses, a study was conducted to collect data from a panel of small business owners and managers through Qualtrics. Specific questionnaire items were used as filters to obtain qualified participants; namely, the size of the business (50 employees or less), and ensuring that respondents were responsible for the firm's digital marketing. The data collection process produced 250 usable questionnaires. The businesses represented all had fewer than 50 employees with the majority (82 per cent) having 10 or fewer employees. The sample comprised 71.2 per cent women and 28.8 per cent men; most respondents owned the small business (84 per cent), with the remainder employees or managers who had decision-making responsibilities for marketing (16 per cent). Next, 57.2 per cent of respondents reported being with their organization for five years or less (see Table I for a summary of the demographic characteristics).

Questions assessing whether or not respondents engaged in specific forms of digital marketing found that they used an average of 3.86 of the 11 different categories provided. All respondents used at least one form of digital marketing (range from 1 to 10). A Facebook page was the most frequently used digital marketing activity (181 of 250

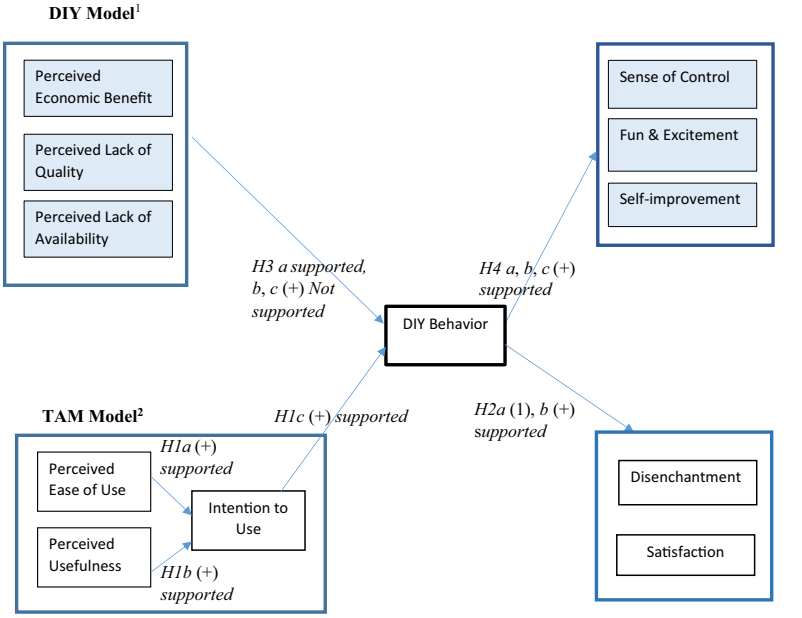


Figure 1.
DIY and TAM digital marketing model

Variable	No.	(%)	Cumulative (%)
Age			
18-30	57	22.8	22.8
31-40	61	24.4	47.2
41-50	60	24	71.2
51-60	49	19.6	90.8
61-up	23	9.2	100.0
Education			
High school diploma/GED	28	11.2	11.2
Some college	56	22.3	33.6
2 year degree	36	14.3	48.0
4 year degree	100	39.8	88.0
Master's/Professional degree	26	10.4	98.4
Doctorate degree	4	1.6	100
Tenure at current firm			
0-5 years	143	57.2	57.2
6-10 years	54	21.6	78.8
>10 years	53	21.2	100
Annual income			
$12,000 or less	13	5.2	5.2
$12,001-40,000	49	19.6	24.8
$40,001-50,000	51	20.4	45.2
$50,001-70,000	57	22.8	68.0
$70,001-90,000	37	14.8	82.8
$90,001-100,000	19	7.6	90.4
$100,001 or more	24	9.6	100

Table I. Participant demographics ($N = 250$)

respondents), followed by a company website (154 respondents), email (106 respondents) and Twitter (79 respondents; although 83 respondents indicated that they used other forms of social media) (Table II).

Measures

We used existing scales to measure the constructs contained in the model. The DIY motivations and outcomes scales are adapted from Wolf and McQuitty (2013), and the

Digital marketing activity	No. participating ($N = 250$)	% using
Facebook page	181	72.4
Website	154	61.6
Email	106	42.4
Other social media	83	33.2
Twitter	79	31.6
SEO	79	31.6
Blog content	75	30.0
Review analytics	66	26.4
e-commerce site	56	22.4
Mobile website	52	20.8
YouTube channel	35	14.0

Table II. Respondent participation in digital marketing activities

TAM-related scales are adapted from Davis (1989). Items in most of the scales use a seven-point Likert scale format (with 1 = strongly disagree and 7 = strongly agree) for responses, with the exception of the five-item measure of intentions from Kleijnen *et al.* (2007) that uses semantic differential pairings of unlikely-likely, improbable-probable, impossible-possible, uncertain-certain and definitely would not use-definitely would use; and the DIY behavior scale, which is a multiple-act-criterion scale (Epstein, 1980; Lastovicka and Joachimsthaler, 1988) sourced from Wolf and McQuitty (2013). The scales and items used in the study appear in the Appendix.

The psychometric properties of these scales were assessed and the Cronbach's alphas range from 0.789 to 0.917 (Table III), which indicates that the reliabilities for the constructs are high (and provide evidence for convergent validity). The average variance extracted (AVE) values for the scales exceed 50 per cent, and are greater than the squared correlations between the constructs (construct correlations range from 0.01 to 0.70); these figures provide evidence of discriminant validity for the different constructs.

Structural equation model

The relationships and hypotheses shown in Figure 1 were tested using a structural equation model with LISREL 8.80. A covariance matrix and maximum likelihood estimation were used to estimate model parameters, and missing data were handled with pairwise deletion. The structural model combines two existing models (the TAM and DIY behavior models), and adds satisfaction with digital marketing activity and intentions to discontinue digital marketing activity as further dependent constructs. Thus, there are 12 constructs in the model: 3 from the TAM (perceived usefulness, ease of use and intentions to use); 7 from the DIY model, including the 3 motivations for DIY behavior (economic, lack of quality and lack of availability), the 3 DIY outcomes (a sense of control in life, fun and excitement and a sense of self-improvement), and the DIY behavior construct reflecting the use of digital marketing; and 2 outcomes of digital marketing (satisfaction with digital marketing and intentions to discontinue digital marketing).

Despite a large model with 12 constructs and 51 observed items, the model estimation converged with no warnings and produced the following goodness-of-fit statistics: $\chi^2(1,103) = 2{,}054.80$ (P = 0.00), CFI = 0.96, NNFI = 0.96, SRMR = 0.080 and

Construct	No. of items	Alpha	AVE
Economic benefit	6	0.890	0.579
Lack of product availability	5	0.907	0.685
Lack of product quality	5	0.915	0.700
Control	4	0.861	0.621
Fun and excitement	4	0.901	0.701
	4	0.789	0.514
Self-improvement			
Satisfaction	4	0.907	0.716
Intentions to discontinue digital marketing	4	0.830	0.615
TAM ease	5	0.917	0.694
TAM usefulness	8	0.914	0.588
DIY behavior*	1		
Intentions to use digital marketing*	1		

Note: *DIY behavior and intentions to use digital marketing are each measured by one summed scale

Table III. TAM and DIY behavior model constructs (number of scale items, scale reliabilities and average variance extracted)

RMSEA = 0.059 (with a 90 per cent confidence interval of 0.055-0.063). The χ^2/df ratio < 2.0, and the model's fit to the data is interpreted as good on the basis of these fit statistics, particularly in light of the statistical power associated with the RMSEA statistic approaching 1.0 (test of close fit, MacCallum *et al.*, 1996), so the goodness-of-fit statistics are assumed conservative (Kaplan, 1995; McQuitty, 2004). Due to the large number of items and constructs, the modification indices suggest many additional paths and error covariances; however, only three within-construct error covariances are estimated to capture correlations between items that are not fully explained by their common factor, and one item was dropped from the analysis due to a significant cross-loading (the item "I design Internet promotions because Internet marketing professionals often do not offer what I want" loaded on the *lack of availability* and the *lack of quality* constructs).

Results

We use the structural equation model's path coefficients to evaluate the hypotheses, and the results are summarized in Table IV. The TAM portion of the model finds that ease of digital marketing use (*H1a*, $t = 5.84$) and perceived usefulness (*H1b*, $t = 2.27$) are significantly related to intentions to adopt digital marketing. The relationship between intentions to adopt digital marketing and actual digital marketing behavior also is significant (*H1c*, $t = 3.75$). As hypothesized, digital marketing behavior is negatively related to intentions to discontinue (*H2a*, $t = -3.73$) and positively related to satisfaction (*H2b*, $t = 4.07$), and both of these relationships are significant.

The DIY behavior model's relationships are not all significant. The relationships between digital marketing behavior and the three DIY motivators (economic, lack of quality, and lack of availability) finds that the perceived economic benefit is significantly related to digital marketing behavior (*H3a*, $t = 2.52$). The perceived lack of availability of digital marketing services also is related to digital marketing behavior (*H3c*, not quite significant with $t = 1.84$), but a perceived lack of digital marketing quality is unrelated to undertaking digital marketing behavior (*H3b*, with $t = 0.22$). All three DIY outcomes (a sense of control, fun and excitement and self-improvement; *H4a*, *H4b* and *H4c*, respectively) are significantly related to digital marketing behavior (with $p < 0.01$).

Hypothesis	Standardized structural coefficients	*t*-statistic*	*p*-value
H1a: Ease of Use → Intentions to Use	0.68	5.84	<0.01
H1b: Usefulness → Intentions to Use	0.21	2.27	<0.05
H1c: Intentions to Use → DIY Behavior	0.73	3.75	<0.01
H2a: DIY Behavior → Intentions to Discontinue	−0.45	−3.73	<0.01
H2b: DIY Behavior → Satisfaction	0.63	4.07	<0.01
H3a: Economic Benefits → DIY Behavior	0.20	2.52	<0.05
H3b: Lack of Product Quality → DIY Behavior	0.002	0.02	NS
H3c: Lack of Product Availability → DIY Behavior	0.19	1.84	NS
H4a: DIY Behavior → Control	0.72	4.15	<0.01
H4b: DIY Behavior → Fun	0.79	4.30	<0.01
H4c: DIY Behavior → Self-improvement	0.81	4.07	<0.01

Notes: *t_{crit} for $p < 0.01$ is 2.58; for $p < 0.05$ is 1.96

Table IV. Summary of hypothesis tests

Overall, the structural equation model's goodness of fit statistics suggest that the model cannot be rejected based on the data, and it appears that the TAM (Davis, 1989) is a good fit-both conceptually and empirically-with the DIY behavior model (Wolf and McQuitty, 2013). Moreover, there is support for the relationships between the two models: specifically, DIY behavior (digital marketing activity) is positively related to intentions to use technology and satisfaction with digital marketing, and negatively related to intentions to discontinue digital marketing.

Discussion and conclusions

The goal of this research is to investigate the antecedents and outcomes of behaviors associated with small businesses' use of digital marketing. Small businesses are less likely to participate in digital marketing than larger businesses, but the majority of the digital marketing literature explores the behavior of large organizations and leaves the reasons for the limited adoption and expected outcomes of digital marketing by small businesses mostly unexplored (Celuch and Murphy, 2010; Järvinen *et al.*, 2012). We use TAM (Davis, 1989) and the DIY behavior model (Wolf and McQuitty, 2013) to study the digital technology adoption behaviors of small business owners and managers. The TAM uses perceptions of the ease of use and the usefulness of technology to explain intentions to use a technology. The DIY behavior model considers the motivations for and outcomes of DIY behaviors. However, if the various activities that small business owners and managers undertake to create value for their firms are viewed as DIY behaviors, then the DIY model is applicable in the small business context.

Where the TAM directly considers the perceived benefits of a technology, the DIY behavior model incorporates marketplace characteristics (perceived economic benefits, perceived lack of quality and perceived lack of availability) that are capable of influencing decisions about implementing digital marketing strategies. The DIY behavior model's outcome variables (senses of control, fun and excitement, and self-improvement) provide insights about the effects of developing digital marketing as a DIY behavior. The conceptual model also considers the TAM's post-adoption factors with the relationships between satisfaction with digital marketing and intentions to discontinue use of the technology to DIY behaviors.

The results from a structural equation model and the hypothesis tests suggest that our sample of small business owners and managers is motivated to undertake digital marketing themselves because of the economic benefits, but not due to a perceived lack of quality. The perceived lack of availability of digital marketing options is positively related to digital marketing behavior, yet the relationship is not quite significant (with $t = 1.84$). All other hypothesized relationships are significant, and the ease of use and usefulness are related to intentions to use digital marketing, which is in turn related to DIY behavior in the small business digital marketing context. DIY behaviors are related to the three DIY outcomes (senses of control, fun and excitement, and self-improvement), and also the TAM derived satisfaction with digital marketing and intentions to discontinue use (negatively related to the latter construct, as hypothesized). We discuss the implications of these findings for marketing theory and for managers, and then explore any limitations of the study and ideas for future research.

Implications for research and practice

This research contributes to the knowledge and understanding of small business owners and managers' digital marketing adoption behavior in several ways. We integrate two existing models that have different explanations of owner/managers' adoption of digital marketing in the small business context. Empirical validation is provided for the TAM with

respect to small businesses' adoption of digital marketing. Although the TAM has been applied to both small and large firms, by integrating the DIY behavior model (Wolf and McQuitty, 2013) we find support for the idea that the technological benefits may not be the only effective motivators for small business owner/managers who decide to undertake digital marketing on their own, and alternative motivations can be important to the acceptance and implementation of digital technologies.

Other implications arise from the finding that, despite its origin in the home improvement industry, the DIY behavior model appears to apply not only to consumers who undertake their own projects, but also to small business owner/managers who perform tasks that require developing specialized knowledge. This finding has implications for both DIY and small business research. Similar to Wolf and McQuitty (2013), we find evidence that marketplace circumstances can affect small business owners and managers' decisions to purchase or self-produce the various forms of expertise required to run their companies, such as digital marketing. Moreover, the decision complexity for small businesses may be more closely related to individuals who perform DIY activities than large corporations, in terms of budgetary constraints, personal interest in the project outcome, control over the process, customization of the product itself, and the personal outcomes derived.

Businesses with less than 50 employees account for almost 60 per cent of global GDP, and their employee count equals that of larger corporations (Kushnir *et al.*, 2010). Small business owner/managers who undertake digital marketing themselves represent lost sales and competition to digital media providers who are not in touch with the small business segment. The implications of the combined DIY-TAM combined model should act as a wake-up call to the digital media industry, because neglecting small businesses' digital marketing needs could mean losing profitable interactions with a powerful economic segment. This is particularly relevant because small businesses, like consumers, typically operate under financial constraints and likely find that completing a DIY project allows the firm to free resources for other projects.

Another implication deriving from the study results is that small business owners and managers who engage in digital marketing activities experience senses of control, fun and excitement, and self-improvement. These DIY outcomes reflect higher-order benefits arising from "direct participation in the process of planning, designing, and construction" (Wolf and McQuitty, 2013, p. 198) of digital marketing. Managers who experience such feelings likely are motivated to continue performing digital marketing themselves, and this observation could apply to other small business related DIY activities. Moreover, a happy and successful owner/manager should contribute to the success of the small business and other employees (Nair and Rao, 2016).

Small business owners and managers invest time and effort to foster loyal customers through personalized experiences. Having a website or digital storefront enables owner/managers to participate in two-way communications with current and prospective customers more frequently and efficiently than other forms of marketing media. In addition to tools such as Google Analytics, small business owner/managers can judge for themselves the effectiveness of digital marketing using measures such as the ratio of positive over negative online customer recommendations and reviews, their quality, and other forms of customer participation. Such feedback can be useful for improving a small business' offerings, which can further improve customer satisfaction and loyalty and reduce the digital divide between small and large businesses.

Third party digital marketing firms that serve small businesses could design cost effective tools, perhaps with packages based on what the small business actually uses. Such firms also could focus on individualized outcome benefits for small business owners (e.g.

more control over customization, certifications for achieving learning thresholds, or tutorials to develop creativity). Promises to build your brand, "seize 30 per cent share of voice from global competitors" (hootsuite.com), or to "program lead generation and management software including capture, scoring, distribution, nurturing, analytics, and more solutions" (Chetu.com) likely will not appeal to the small business owner/manager, because they do not support owners and managers who prefer to undertake their own digital marketing. Consequently, an alternative approach for third party digital marketing firms serving small businesses could be, for example, an invitation to owners and managers to learn a new skill for engaging with their current customers and gaining new customers using mobile devices. Tutorials could guide the small business owner or manager through the process with a template and an easy conversion to live digital marketing. As a real life example, Constant Contact's email marketing provides a real world approach that could resonate with small business marketers: "Send great-looking emails that drive big results" (www.constantcontact.com). Similarly, templates and drag and drop functions could increase the appeal for DIY marketers.

Limitations and future research

A limitation to our study is that we focus only on intentions and marketplace motivations (perceived economic benefits, lack of quality, and lack of availability) to explain digital marketing activities, but alternative motivations can exist. For example, small business owners and managers also could be motivated to perform digital marketing themselves simply to obtain a customized product (Wolf and McQuitty, 2011). Further study of DIY behaviors in the small business digital marketing context is needed to evaluate whether other reasons for such behaviors exist (this limitation and research opportunity also exists for the outcomes of DIY behaviors). Another possible limitation to our study is that the sample may not be representative of all owners and managers who perform digital marketing for their small businesses.

Our study also suggests ideas for future research. For example, for our sample, the perceptions of the ease of use associated with digital technologies was more strongly related to the small businesses owners' and managers' digital marketing behaviors than the perceived usefulness of the technology. This difference may indicate that some owners and managers are not convinced the technology yields the returns promised by advertisers. Further research is needed to study whether there are ways of making digital marketing either easier to use or at least appear easy to use.

Combining the TAM and DIY behavior models to explain small business owners' and managers' digital marketing activities sheds some light on the characteristics of the small business digital footprint. However, more research is needed to identify optimal environments in which small business owners and managers increase digital marketing adoption and close the digital gap that exists with large corporations. Another idea is exploring alternative motivations for the owners/managers who undertake digital marketing activities themselves, such as higher order constructs such as technology readiness (Parasuraman and Colby, 2015) and trust (Pavlou, 2003; Skard and Nysveen, 2016). Technology readiness can affect an individual's predisposition to use new technologies through motivators (optimism and innovativeness) and inhibitors (discomfort, and insecurity) (Parasuraman and Colby, 2015). Trust can be integrated into the TAM through trust in the vendor and trust in the technology (Pavlou, 2003; Skard and Nysveen, 2016). Additional constructs such as perceived reliability, ability, and ethics also could be a part of the framework.

The pattern of motivations for DIY behaviors in our study is very similar to those from Wolf and McQuitty (2013), with the exception that the perceived lack of available product options was significant in that study. The perceived lack of availability of digital marketing options is related to digital marketing behavior, but the relationship is not quite significant (with $t = 1.84$). The difference could be attributable to the perception that options exist for digital marketing services, but more study is needed to validate these findings.

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Appendix. Technology acceptance model and do-it-yourself behavior model scales

Technology acceptance model constructs

Perceived Usefulness Scale (adapted from Davis, 1989).

(1) Using the Internet to promote our products or services for my business would enable the company to accomplish growth more quickly.

(2) Using the Internet to promote our products or services would improve business performance.

(3) Using the Internet to promote our products or services for my business would increase our productivity. Using the Internet to promote our products or services would enhance the company's effectiveness to increase awareness of the business.

(4) Using the Internet to promote our products or services would enhance the company's effectiveness to increase customer engagement in the business.

(5) Using the Internet to promote our products or services would enhance the company's effectiveness to increase lead generation for the business.

(6) Using the Internet to promote our products or services would make it easier to run the business.

(7) I believe having Internet promotions our products or service would be useful for the business.

Perceived ease of use scale (adapted from Davis, 1989).

(1) Learning to create promotions on the Internet for our products or services would be easy for me.

(2) I would find it easy to achieve Internet promotion objectives.

(3) My interaction with promoting products or services on the Internet would be clear and understandable.

(4) It would be easy for me to become skillful at Internet promotions of our products or services.

(5) I find developing Internet promotions for our products or services easy.

Intention scale (adapted from Kleijnen *et al.*, 2007).

(1) Impossible – Possible

(2) Uncertain – Certain

(3) Definitely would not use-definitely would use

(4) Improbable – Probable

(5) Unlikely – Likely

Satisfaction Continuance scale (adapted from Sun, 2013).

(1) "Extremely displeased"___ 4 "Neutral"____7 "Extremely pleased"

(2) "Extremely frustrated" ___ 4 "Neutral" ___7 "Extremely content"

(3) "Extremely terrible" ____ 4 "Neutral"____7 "Extremely delighted"

(4) "Extremely dissatisfied" ___4 "Neutral"____7 "Extremely satisfied"

Intention to discontinue scale (adapted from Sun, 2013).

(1) I intend to discontinue my use online marketing even though I am not particularly dissatisfied with it because I found another technology that is superior for my needs.

(2) I predict that I will not use online marketing any longer, even if I cannot find something else to replace it, because it does not fit my needs.

(3) I plan to stop using online marketing, and to find something else because I am dissatisfied with it.

(4) I plan to stop using Facebook, using something else superior instead.

Do-it-yourself behavior model constructs

DIY Behavior scale (adapted from Wolf and McQuitty, 2013).

Please indicate which online marketing activities you perform for your company:

(1) Email

(2) Create or post blog content

(3) Create or post website content

(4) Create or post Facebook content

(5) Create or post mobile content

(6) Create or post YouTube content
(7) Create Search Engine Optimization key words
(8) Review online analytics
(9) Create or post Twitter content
(10) Create or update content on other social media site(s)
(11) Manage and update eCommerce site

Perceived economic benefit scale (adapted from Wolf and McQuitty, 2013).

(1) I do my own Internet promotion of our products/services to save money.
(2) Performing my own maintenance on the Internet promotions saves me money.
(3) By managing the Internet promotions for my business, I can spend money on other things.
(4) I find that I can save a lot of money by managing the Internet promotions of our products/services myself.
(5) When I create an Internet promotion myself, the money I can save is important.
(6) When I begin an Internet promotion project I expect to save money.

Perceived lack of quality scale (adapted from Wolf and McQuitty, 2013).

(1) Designing your own Internet promotions is good because website developers are unreliable.
(2) Designing your own Internet promotions makes sense because Internet marketing professionals do not do what I want.
(3) Designing your own Internet promotions is good because I can do a better job than the professional website developers.
(4) Hiring an Internet marketing professional results in worse work than when I do it myself.
(5) The work of people I can hire is not of good quality so I have to do the work myself.

Perceived lack of product availability scale (adapted from Wolf and McQuitty, 2013).

(1) I will create my own Internet promotions to better match my business needs.
(2) I design Internet promotions because Internet marketing professionals often do not offer what I want.
(3) To get the customized digital content I need, I have to create it myself.
(4) To get an Internet promotion for our products/services that is compatible with my business, I have to do things myself.
(5) To get an Internet promotion for our products/services that is consistent with my marketing strategy, I have to make it myself.

Sense of control scale (adapted from Wolf and McQuitty, 2013).

(1) Do-It-Yourself (DIY) online marketing projects help a person manage business goals.
(2) Do-It-Yourself (DIY) online marketing projects help a person better control the project.
(3) Do-It-Yourself (DIY) online marketing projects help a person be in charge of the project situation.
(4) Do-It-Yourself (DIY) online marketing projects gives a person confidence over the project situation.

Sense of fun and excitement scale (adapted from Wolf and McQuitty, 2013).

(1) DIY online marketing projects can be exciting.
(2) DIY online marketing projects can be entertaining.
(3) DIY online marketing projects can be interesting.
(4) DIY online marketing projects can be enjoyable.

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